

★ WEEK 10 · PART 1 · PRACTICE

Markov Regimes, Worked by Hand

Persistence and the Long Run; Which Regime Made This Return

The transition matrix and the emissions, made concrete

Part 1's foundations, with the arithmetic shown. We read a regime transition matrix for duration and long-run mix, then use emission distributions and one Bayes step to infer which hidden regime a return most likely came from.

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PART 1 · APPLICATIONS

Two Regime Calculations, by Hand

Case A: a calm/turbulent transition matrix, worked for expected duration, a two-step probability, and the long-run mix. Case B: emission distributions and one Bayes step to infer the hidden regime from a return. Then a Claude Code lab.

What Part 1 Told Us

Now we make it concrete - with numbers

1

A Markov chain has one-step memory

The next regime depends only on the current one. All the switching probabilities live in a transition matrix.

2

Regimes are sticky, and settle

High stay-probabilities make regimes persist; over the long run the chain spends a fixed fraction of time in each.

3

The state is hidden; emissions reveal it

Each regime has its own return distribution. A return is a clue - and one Bayes step turns it into a regime probability.

Everything here is a calculator exercise: one transition matrix, two return distributions, one bad day.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from the matrix or the distributions.
- Compute duration, mix, or a posterior by hand.
- Then read what it says about the market.

What you should watch

- How stickiness sets the expected duration.
- How the chain settles to a long-run mix.
- How one return shifts the regime belief.

Reading the Transition Matrix

Persistence, and where the chain settles

Two regimes: a calm market and a turbulent one. The transition matrix holds the switching probabilities. From it alone - no data yet - we can read how long each regime lasts, the chance of switching over two steps, and the fraction of time the market spends turbulent in the long run

The matrix

- Calm stays calm 95% of days.
- Turbulent stays turbulent 80%.
- Calm is stickier than turbulence.

What we'll do by hand

- Expected duration of each regime.
- The two-step transition probability.
- The long-run stationary mix.

The Transition Matrix

All the switching in one grid

$$P = \begin{pmatrix} 0.95 & 0.05 \\ 0.20 & 0.80 \end{pmatrix} \quad \begin{array}{l} \text{row} = \text{from Calm / Turbulent} \\ \text{col} = \text{to Calm / Turbulent} \end{array}$$

How to read it

- Each row is where you are now.
- Each column is where you go next.
- Rows sum to one - you go somewhere.

What it encodes

- The diagonal is the stay-probability.
- The off-diagonal is the switch chance.
- That's the whole dynamics of the regimes.

Expected Duration

How long a regime lasts

$$\text{expected duration} = \frac{1}{1-p_{\text{stay}}} : \quad \text{Calm} = \frac{1}{1-0.95} = \mathbf{20} \text{ days}, \quad \text{Turbulent} = \frac{1}{1-0.80} = \mathbf{5} \text{ days}$$

The formula

- Expected duration = $1 / (1 - \text{stay probability})$.
- It's the mean of a geometric distribution.
- Higher stickiness, longer spells.

The numbers

- Calm lasts about 20 days on average.
- Turbulence lasts about 5 days.
- Calm periods are four times longer.

A Two-Step Transition

Probabilities compound

$$P(\text{Calm} \rightarrow \text{Calm in 2}) = 0.95(0.95) + 0.05(0.20) = \mathbf{0.9125}$$

$$P(\text{Calm} \rightarrow \text{Turb in 2}) = 0.95(0.05) + 0.05(0.80) = 0.0875$$

To be calm in two days you can stay calm both days, or dip into turbulence and come back. Summing the paths gives 0.9125 - the entries of the squared matrix.

The Long-Run Mix

Where the chain settles

$$\pi_C (0.05) = \pi_T (0.20), \quad \pi_C + \pi_T = 1 \Rightarrow \pi_C = \mathbf{0.80}, \quad \pi_T = \mathbf{0.20} \text{ (long-run mix)}$$

The balance condition

- In steady state, flow out of calm equals flow in.
- So π_C times 0.05 equals π_T times 0.20.
- With the two summing to one.

The answer

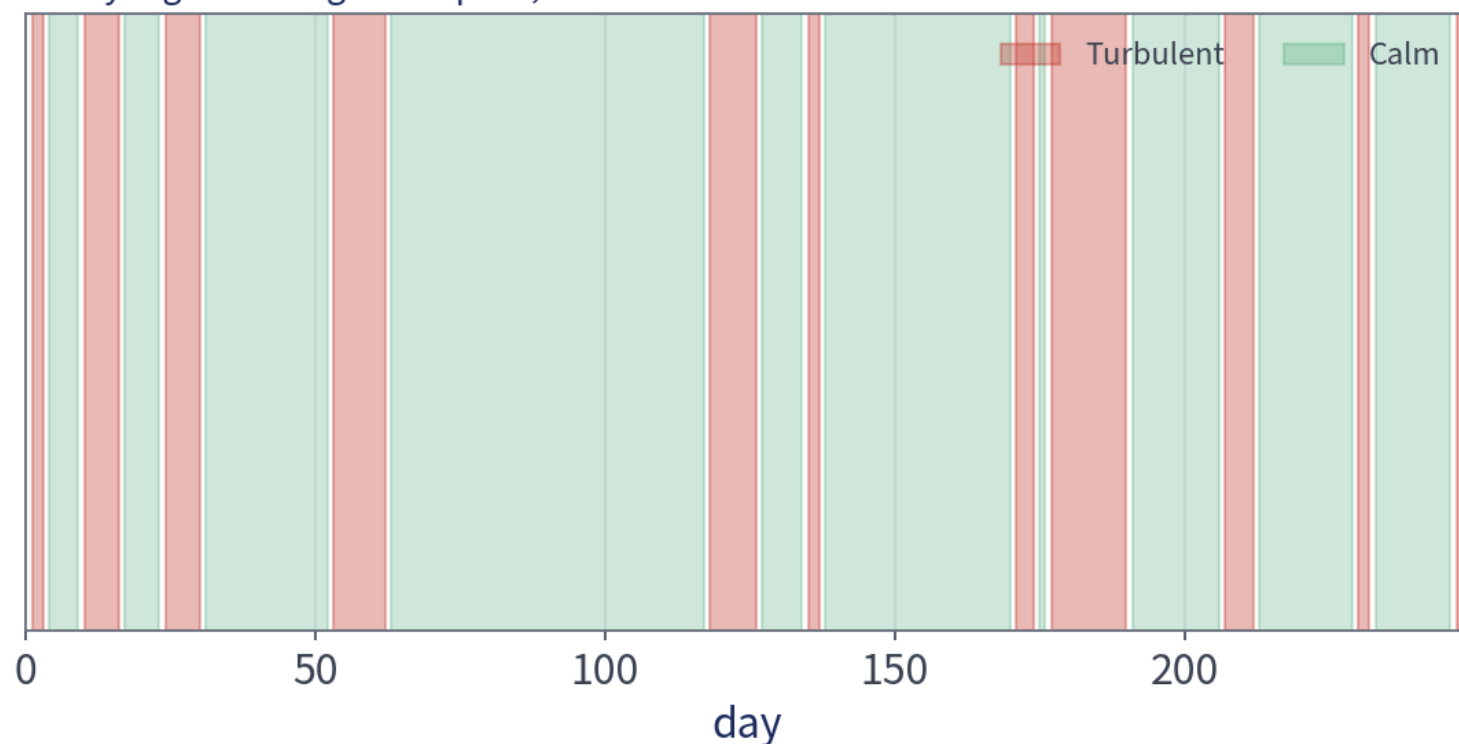
- The market is calm 80% of the time,
- and turbulent 20%.
- Independent of where it starts.

Sticky Regimes, a Steady Mix

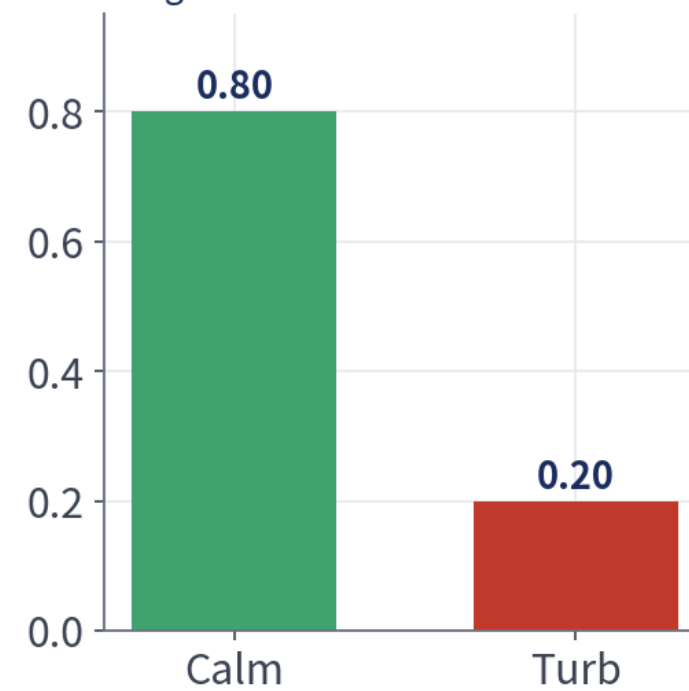
A simulated path shows the signature of sticky regimes: long green calm stretches broken by short red turbulent bursts. Over the long run, the fractions settle exactly to the stationary mix - 80% calm, 20% turbulent - no matter where the chain began

Case A: 80% calm / 20% turbulent in the long run - calm lasts 20 days, turbulence 5

Sticky regimes: long calm spells, short turbulent bursts



Long-run mix



What the Matrix Tells You

Structure, before any data

You already know

- How long a calm or turbulent spell tends to last.
- The odds of switching over any horizon.
- The long-run share of turbulent days.

Why it matters

- It sets expectations for regime persistence.
- It calibrates how fast to react to a switch.
- It's the backbone the emissions hang on.

Which Regime Made This Return?

Emissions, and one Bayes step

The regime is hidden - but each one emits returns with its own distribution. Calm returns are small and positive; turbulent returns are negative and wild. Given a -2% day, we compute how likely that return is under each regime, then combine with the prior to get the probability we're now in the turbulent regime

The emissions

- Calm: returns $\sim N(+0.5\%, 1\%)$.
- Turbulent: returns $\sim N(-1\%, 3\%)$.
- Today's return is -2.0%.

What we'll do by hand

- Compute the likelihood under each regime.
- Multiply by the prior regime probabilities.
- Normalize to a posterior.

The Likelihood of the Return

How well each regime explains -2%

$$\text{Calm} \sim N(+0.5, 1.0^2) : \ell_C = \varphi\left(\frac{-2.0-0.5}{1.0}\right) = \varphi(-2.5) = 0.018$$

$$\text{Turb} \sim N(-1.0, 3.0^2) : \ell_T = \varphi\left(\frac{-2.0+1.0}{3.0}\right) = \varphi(-0.33) = 0.126$$

A -2% return is 2.5 standard deviations below the calm mean but well within the turbulent spread - so its density is about seven times higher under turbulence.

One Bayes Step

From likelihood to regime probability

$$P(\text{Turb} \mid r) = \frac{\pi_T \ell_T}{\pi_C \ell_C + \pi_T \ell_T} = \frac{0.20(0.126)}{0.80(0.018) + 0.20(0.126)} = \mathbf{0.64}$$

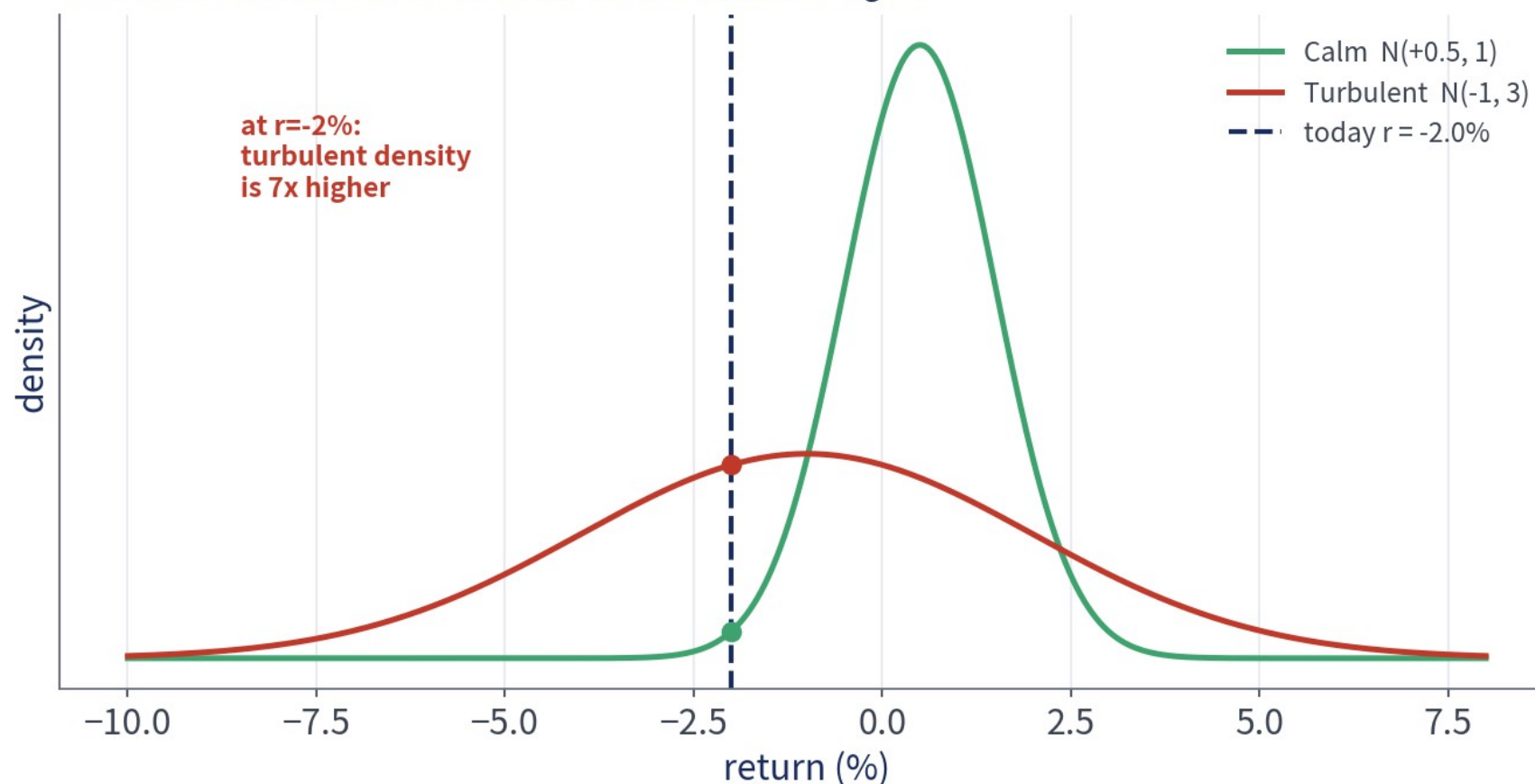
$\Rightarrow$ a -2% day lifts turbulent belief from 0.20 to 0.64

Combining the prior (20% turbulent) with the likelihoods lifts the posterior to 0.64 - the single bad day more than triples our belief that we're in turbulence.

The Return Points to a Regime

At -2%, the calm distribution (green) has almost no density - it's a 2.5-sigma event there - while the turbulent one (red) is comfortably wide. That mismatch in density is what Bayes turns into a posterior, shifting belief sharply toward turbulence

Case B: the -2% return is far likelier under the turbulent regime



Reading the Hidden State

A return is evidence, not proof

What one return gives

- Not certainty - a shifted probability.
- A big loss is stronger evidence than a small one.
- The prior still matters, especially early.

Why this is the seed

- This is the emission half of the HMM.
- Chaining it over time is the forward algorithm.
- That's exactly where Part 2 goes.

The Two Ingredients of an HMM

Transitions and emissions

An HMM is just these two pieces. The transition matrix says how regimes evolve - their stickiness and long-run mix. The emissions say how each regime shows up in returns. Case A worked the transitions; Case B worked the emissions and one inference step.

Together they define the whole model

Transitions (Case A)

- How the hidden regime moves.
- Duration and the long-run mix.
- The backbone of persistence.

Emissions (Case B)

- How each regime generates returns.
- The likelihood of an observation.
- The clue that reveals the hidden state.

Reproduce the Regime Calcs

Direct Claude Code to check your arithmetic

GOAL Verify the transition-matrix reads and the Bayes step, then simulate a regime path.

[INPUT] Prompt · Claude Code

Write and run `part1_practice.py` that:

CASE A - transitions:

- * from $P = \begin{bmatrix} 0.95 & 0.05 \\ 0.20 & 0.80 \end{bmatrix}$, print each regime's expected duration = $1/(1-\text{stay})$;
- * print P squared and the stationary distribution;
- * simulate a path and check the long-run mix.

CASE B - emissions:

- * with $\text{Calm} \sim N(0.5, 1)$ and $\text{Turb} \sim N(-1, 3)$, print the density of $r = -2$ under each;
- * apply Bayes with prior $0.80/0.20 \rightarrow$ posterior.

Match every printed number to the slides.

Check against the slides

- duration 20 and 5 days.
- two-step $C \rightarrow C = 0.9125$.
- stationary $0.80 / 0.20$.
- posterior turbulent = 0.64.

Deliverable: `part1_practice.py` whose printed numbers match the transition and emission calcs.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 How do you get a regime's expected duration?

$1 / (1 - \text{stay probability})$

Q2 What is the stationary distribution?

The long-run fraction of time in each regime

Q3 How does a return update the regime belief?

Bayes: prior times emission likelihood

All three come from the transition matrix and the emission-plus-Bayes step.

Three Pitfalls to Avoid

Where regime reasoning slips

Common mistakes

- "A high stay-probability means it never switches."
- "One bad return proves we're in a crisis."
- "The starting regime sets the long-run mix."

The reality

- It switches - just rarely; duration is $1/(1-\text{stay})$.
- It's evidence, not proof - Bayes gives a probability.
- The stationary mix is independent of the start.

Two Calculations to Remember

Part 1 applications in one screen

1

Read the transition matrix

Stay-probabilities gave expected durations of 20 days (calm) and 5 days (turbulent); the two-step calm-to-calm probability was 0.9125; and the long-run mix settled at 80% calm, 20% turbulent - all from the matrix alone.

2

Infer the hidden regime from a return

A -2% day had density 0.018 under calm but 0.126 under turbulence. One Bayes step, starting from a 20% prior, lifted the posterior probability of turbulence to 0.64 - the return shifted belief, but didn't prove it.

3

Transitions plus emissions define the HMM

Case A is how the regime moves; Case B is how it shows up in returns. Chaining the emission step over time is the forward algorithm - exactly where Part 2 begins.

Next — Part 2 practice: the forward algorithm, Viterbi, and Baum-Welch, run by hand.